GABRIEL BAZALAR

+1 (765) 767-3784 | gbazalar@purdue.edu | www.linkedin.com/in/gabriel-bazalar | https://gbazalar.github.io/

EDUCATION

Purdue University West Lafayette, IN | **Expected May 2026**

Bachelor of Science in ***Electrical Engineering (BSEE)***

School Ambassador, Semester Honors (Fall 2024)

Relevant Coursework: ASIC Design, Microprocessors Interfacing, Digital Design, Signal Analysis, Probabilistic Methods, Feedback-Control Design, Power Electronics, Advanced C Programming, Python for Data Science.

INDUSTRY EXPERIENCE

Undergraduate Researcher, Flex Laboratory West Lafayette, IN | **May 2024 – Present**

- Designed a low-voltage **piezoelectric energy harvesting PCB** using the LTC3588 IC, integrating over 100 ceramic and polymer sensors embedded in concrete structures.
- Performed impedance matching and analysis using an AIM UHF **impedance analyzer** and oscilloscope, ensuring efficient power transfer and consistent analog readings for measurements of concrete structural strength (5 kHz to 1 MHz).

Circuit Design Lab II Teaching Assistant, Purdue University West Lafayette, IN | **January 2024 – August 2024**

- Simulated circuits in **LTSpice** and **KiCad** to demonstrate RLC transient analysis, MOSFET amplifiers, and Op-Amps, fostering a hands-on learning experience among students.
- Collaborated with graduate engineers on revising over **100+ student** breadboard and PCB circuits using Keysight oscilloscopes and function generators achieving an average **15% improvement** in capstone project scores.

Hardware Engineering Intern, Calite Inmobiliaria, Gerencia y Construcción Lima, Perú | **April 2023 – August 2023**

- Tested **IoT controllers** to ensure optimal performance in security cameras, intercoms, and elevator communication systems; deployed Wi-Fi networks using Ubiquity UniFi to enhance system reliability.
- Implemented an augmented reality visualization tool using Unity Engine for 3D models, enhancing collaboration and understanding among more than **50** architects, structural engineers, and project managers.

PROJECTS

Lunar Lander Computer System

- Designed a Lunar Lander simulator for **FPGA**, interfacing a keypad and 7-segment screen for displaying altitude, velocity, and thrust to determine landing success.
- Prototyped it on an **ICE40H8K FPGA** board and performed Static Timing Analysis (STA) using Siemens **Questa Sim**.

Video Game Console

- Emulated the video game Piano Tiles on an **STM32** microprocessor, interfacing EEPROM, SD card reader, audio amplifier, analog joystick, and 7-segment displays for full console experience.
- Utilized the **CMSIS** framework in Platform.IO for direct modification of timer interrupts and microprocessor registers.

Self-Driving Interfacing System

- Developed a system controller using STM32 NUC for **UART** serial communication with a Jetson TX2 ML driving model, outputting physical PID acceleration for motor control, radio frequency for steering, and pressure for pneumatic brakes.

CAN Communication Board

- Implemented a **CAN** communication board with the low power STM32L432KCU6 for sending and receiving battery information and emergency break signal, prioritizing design for manufacturability (DFM) on Purdue's SAE Racing Team.

LEADERSHIP

Latin Autonomous Projects Club Founder (LAPs) **May 2024 – Present**

- **Founded** an engineering club to support Latino students in developing intelligent systems and robotic projects, fostering innovation in Embedded Hardware and Machine Learning.

ECE Student Ambassador (ECEA) **January 2024 – Present**

- **Planned, managed, and represented** the College of Electrical Engineering at events, allocating a semester budget exceeding $8,000 to fund student recruitment, innovative projects, and student-led initiatives.

SKILLS AND INTERESTS

Programming: System Verilog, RISC-V Assembly, C, Python, VS code, MATLAB

CAD Tools: Questa Sim, Synopsys VCS, Altium Designer, LTSpice, KiCad, 3DsMax, Fusion 360

Certifications: IBM – ML with TensorFlow, NVIDIA – AI on Jetson Nano, IPA'D – Advanced 3DsMax

Languages: Spanish (native), English (native)